

Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

> **Definition to Remember**

1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

3. Key Differences

Feature	Hardwired CU	Microprogrammed CU
Control Logic	Implemented physically using logic gates, flip-flops, and decoders.	Implemented using a microprogram stored in Control Memory (ROM).
Flexibility	Not flexible; changes require physical hardware modification.	Flexible; changes can be made by updating the microprogram.
Cost	Generally lower cost for simple designs.	Generally higher cost due to the need for Control Memory and microprogramming.
Development Time	Longer development time due to physical design and prototyping.	Shorter development time as changes can be made in software.
Scalability	Difficult to scale for complex designs.	Easier to scale for complex designs by updating the microprogram.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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